Reputation, Innovation, and Externalities in Venture Capital

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Outline

1. Motivations and contributions

2. Model

3. Equilibrium characterization

4. Innovation spillovers

5. Reputational externality

6. Robust extensions

Motivations

- In the two-sided markets there is incomplete information about participants' *types* $\Rightarrow$ there is room for *reputation* building.

- Examples (of incomplete information):
  - Labor market $\rightsquigarrow$ firm's productivity
  - Educational market $\rightsquigarrow$ university's quality
  - Venture capital market $\rightsquigarrow$ VC's ability

- Contributions of this paper:
  - develops a search and matching model where agents of one side of the market have reputational concerns;
  - studies two instances of market failure in venture capital: positive spillovers, and reputation-deal flow externality.

VCs' ability and reputation (Sorensen 2007)

A = net impact, C = after adjusting for the sorting

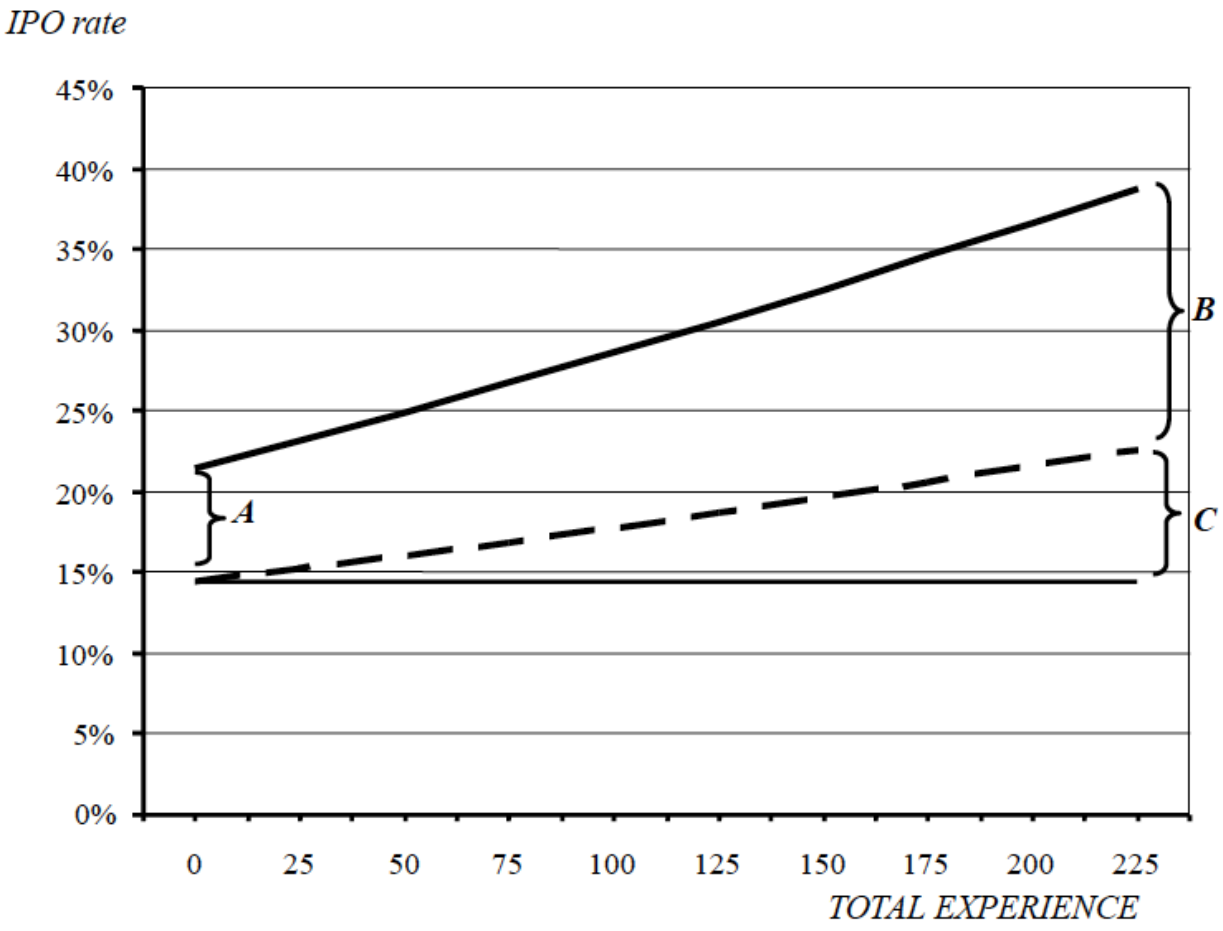

Figure 1: VC's influence disentangled from sorting

Theoretical contributions

- Developing a dynamic equilibrium model of search and matching in an economy where individuals on one side have symmetric incomplete information about their type.
  (Shimer and Smith 2000&2005), (Chade 2005), (Damiano et. al 2005), (Hoppe et. al 2009), (Anderson and Smith 2010), (Anderson 2015), (Chade and Eeckhout 2017)

- Methodological contribution: matching sets $\Rightarrow$ stopping time problems faced by investors $\Rightarrow$ applying the theory of monotonicity to the operators acting on function spaces (Krasnoselskij 1964)

Contributions (to the economics of venture capital)

- In equilibrium VCs earn reputation for their value-added skills, thereby rationalizing a number of stylized facts:
  - Later stage startups attract a wider range of investors (Gompers, Gornall, Kaplan and Strebulaev 2020)
  - More reputable VCs exhibit higher tolerance for failure (Manso 2011), (Tian and Wang 2014)
  - Cost-reducing technological shocks enhances the variety of financed projects: "*spray and pray*" (Ewens, Nanda and Rhodes-Kropf 2018)
- Extending the baseline model to study **two** instances of market failure:
  - *Reputation-deal flow externality* among VCs $\rightsquigarrow$ early termination of startups; sub-optimal mass of active high ability investors
  - *Neglect of positive spillovers* among startups
- Other related literature: (Silveira and Wright 2016), (Akcigit et. al 2019)

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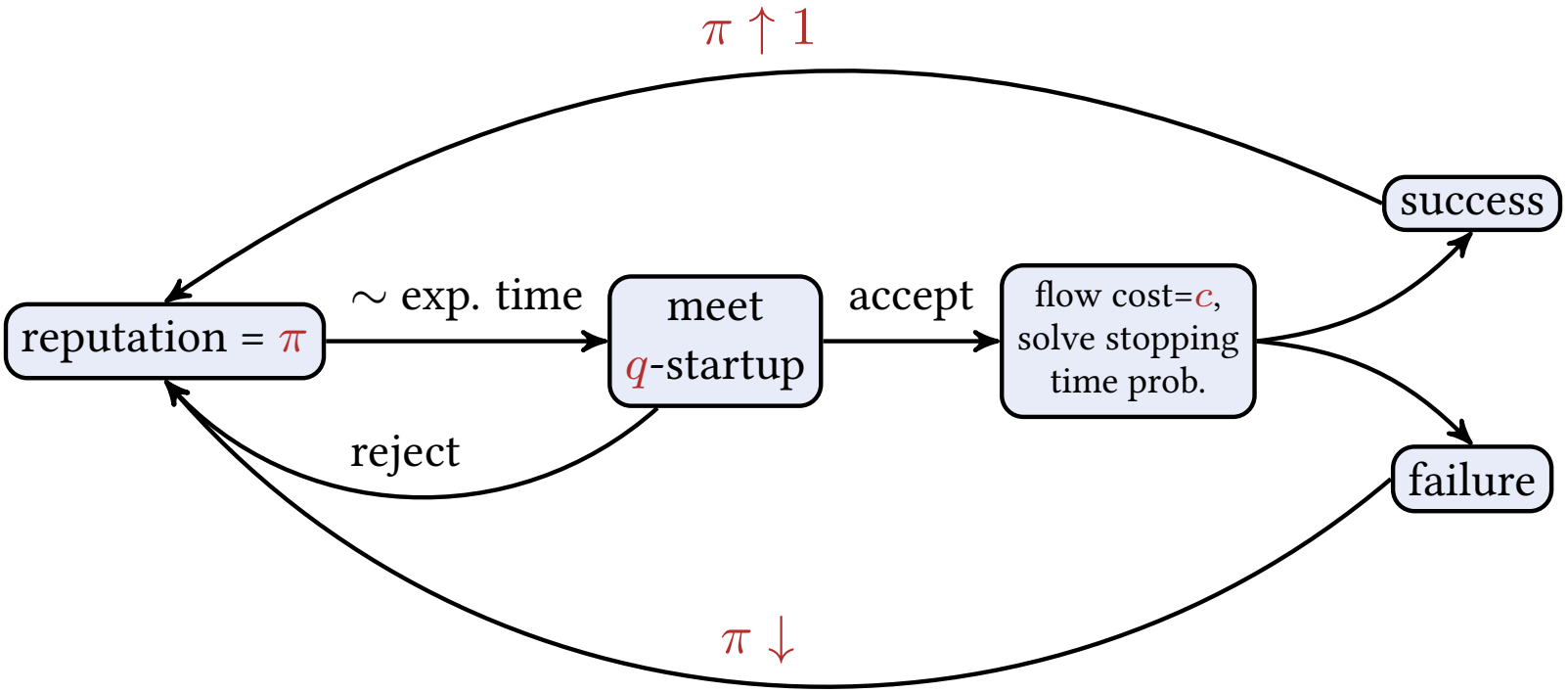
Model diagram

- A unit mass of VCs. A known fraction $\mathfrak{p}$ have high type ($\theta = 1$) while $1 - \mathfrak{p}$ have low type ($\theta = 0$) $\rightsquigarrow$ their types are hidden to EVERYONE.
- Two types of projects $q \in \{a, b\}$ with measures $\varphi_a, \varphi_b \rightsquigarrow$ their types are observable to EVERYONE.
- VCs randomly meet the projects, subject to search friction with quadratic matching technology and frictional rate κ.
- Upon the meeting VC either accepts or rejects the project, thus waiting for the next chance.
- If the partnership is formed
  - VC covers the flow investment cost of c.
  - A successful outcome with unit payoff arrives with intensity $\theta\lambda_q$, where $q \in \{a, b\}$, and $\lambda_b > \lambda_a$.
  - VC collects all the revenues, i.e entrepreneur has zero bargaining power (Ueda 2004) and (Hellman and Puri 2002).
  - VC has the option to terminate the project.

Dynamic timeline



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VC strategy

- Reputation is the market posterior belief about the VC's type $\rightsquigarrow \pi_t := \mathsf{P}\left(\theta = 1 \middle| \mathcal{I}_t\right)$.

- When unmatched the VC's reputation is fixed, and when matched to a q-project:

$$\dot{\pi}_t = -\lambda_q \pi_t (1 - \pi_t) \rightsquigarrow \text{before success}$$
$$\pi_t = 1 \rightsquigarrow \text{after success}$$

- What matters in the equilibrium is the matching set $\mathcal{M}$:

$$(q, \pi) \in \mathcal{M} \Leftrightarrow \text{rep. } \pi \text{ VC finds } q \text{ profitable.}$$

  - this set encodes the decision to accept or reject a project.
  - it also encodes the decision to terminate or continue the funding.

Value functions and the fixed-point

- Matching value function $v(\pi, q)$ non-recursive form :

$$rv(\pi,q) = -c + \lambda_q \pi \underbrace{[1 + w(1) - v(\pi,q)]}_{\substack{\text{surplus upon}\\ \text{success}}} - \underbrace{\lambda_q \pi(1-\pi)\partial_\pi v(\pi,q)}_{\substack{\text{marginal}\\ \text{reputation loss}}}$$

- Reputation value function $w(\pi)$ non-recursive form :

$$rw(\pi) = \kappa\varphi_a \underbrace{[v(\pi,a) - w(\pi)]}_{\substack{\text{surplus gained}\\ \text{from } a\text{-project}}} \chi_a(\pi) + \kappa\varphi_b \underbrace{[v(\pi,b) - w(\pi)]}_{\substack{\text{surplus gained}\\ \text{from } b\text{-project}}} \chi_b(\pi)$$

- Matching set $\mathcal{M} \subset \{a, b\} \times [0, 1]$:

$$\mathcal{M} = \{(q,\pi) : v(\pi,q) > w(\pi)\}$$

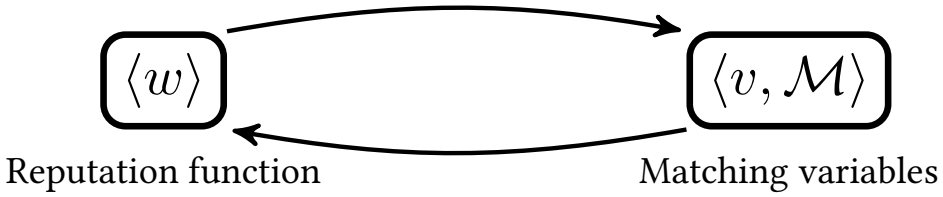


Figure 2: Equilibrium feedbacks

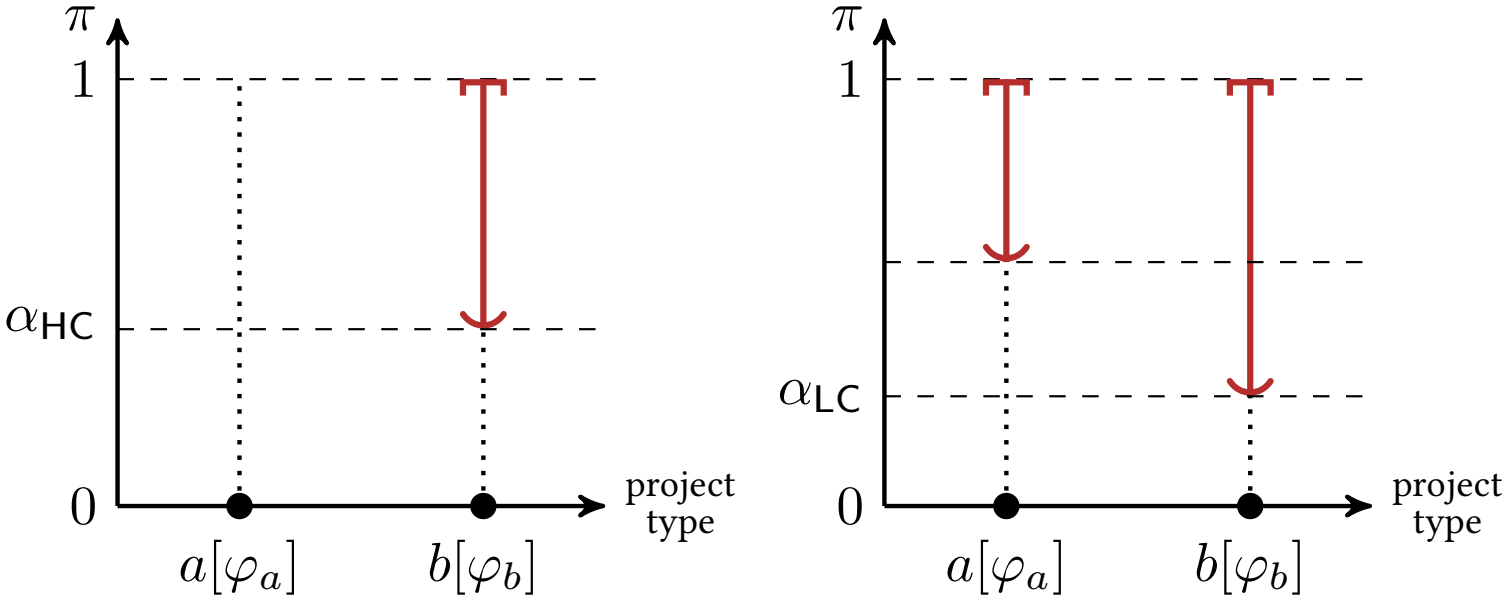
Equilibrium

Theorem 1: Equilibrium existence and uniqueness

$\forall\{\varphi_a, \varphi_b\}$, $\exists$ a stationary equilibrium with increasing value functions in π. This equilibrium is unique for large discount rates.

Regime determination:

$$\lambda_a - c > \frac{\kappa\varphi_b(\lambda_b - c)}{\kappa\varphi_b + r + \lambda_b} = \text{Opportunity cost of forgoing the option to wait}$$



(a) high cost matching sets (b) low cost matching sets

$$\boxed{\alpha = \frac{c}{\lambda_b\,(1 + w(1))}} \rightsquigarrow \text{proxy for imperfect learning}$$

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Spillovers and endogenous supply of startups

- Interpretations of two types of projects ($\lambda_a < \lambda_b$):
  - early vs. late stage startups. $\rightarrow$ time dimension as $\frac{1}{\lambda_a} > \frac{1}{\lambda_b}$
  - radical vs. incremental innovation $\rightarrow$ risk dimension as $\frac{1}{\lambda_a^2} > \frac{1}{\lambda_b^2}$
- Spillovers from early stage small companies $\rightsquigarrow$ late stage businesses.

failed to be internalized in private decisions

- Examples (Mazzucato 2013):
  - iPhone depends on the Internet; the progenitor of the Internet was *ARPANET*: a program funded by DARPA in 1960's.
  - Google maps depends on GPS: a US military project called *NAVSTAR* in 1970's.

A framework to endogenize

- A fraction $\zeta < 1$ of successful early stage projects would translate into the late stage projects:

$$\underbrace{\kappa\varphi_b n(1)}_{\substack{\text{outflow from}\\ \text{late stage}}} = \underbrace{\zeta\lambda_a m_a(1)\chi_a(1)}_{\substack{\text{inflow to}\\ \text{late stage}}}$$

$$\varphi_b = \begin{cases} \zeta\varphi_a & \chi_a(1) = 1 \\ 0 & \text{o.w} \end{cases}$$

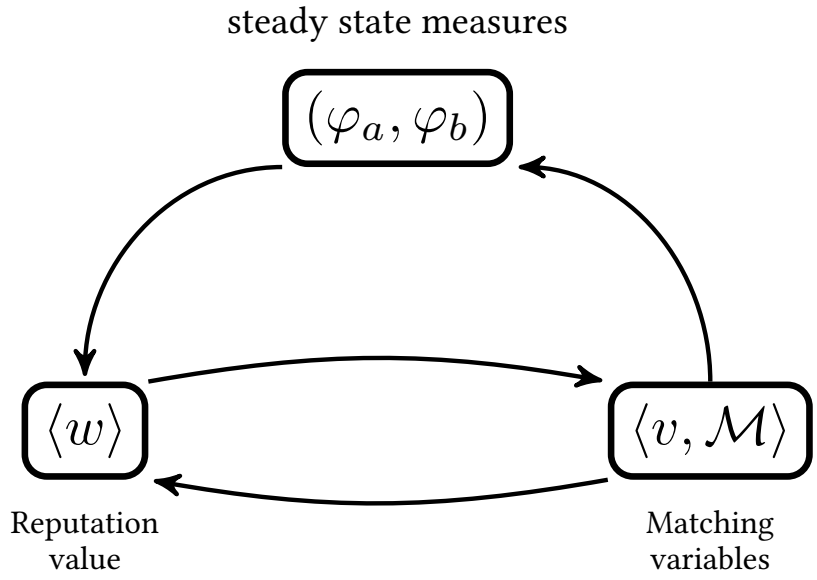


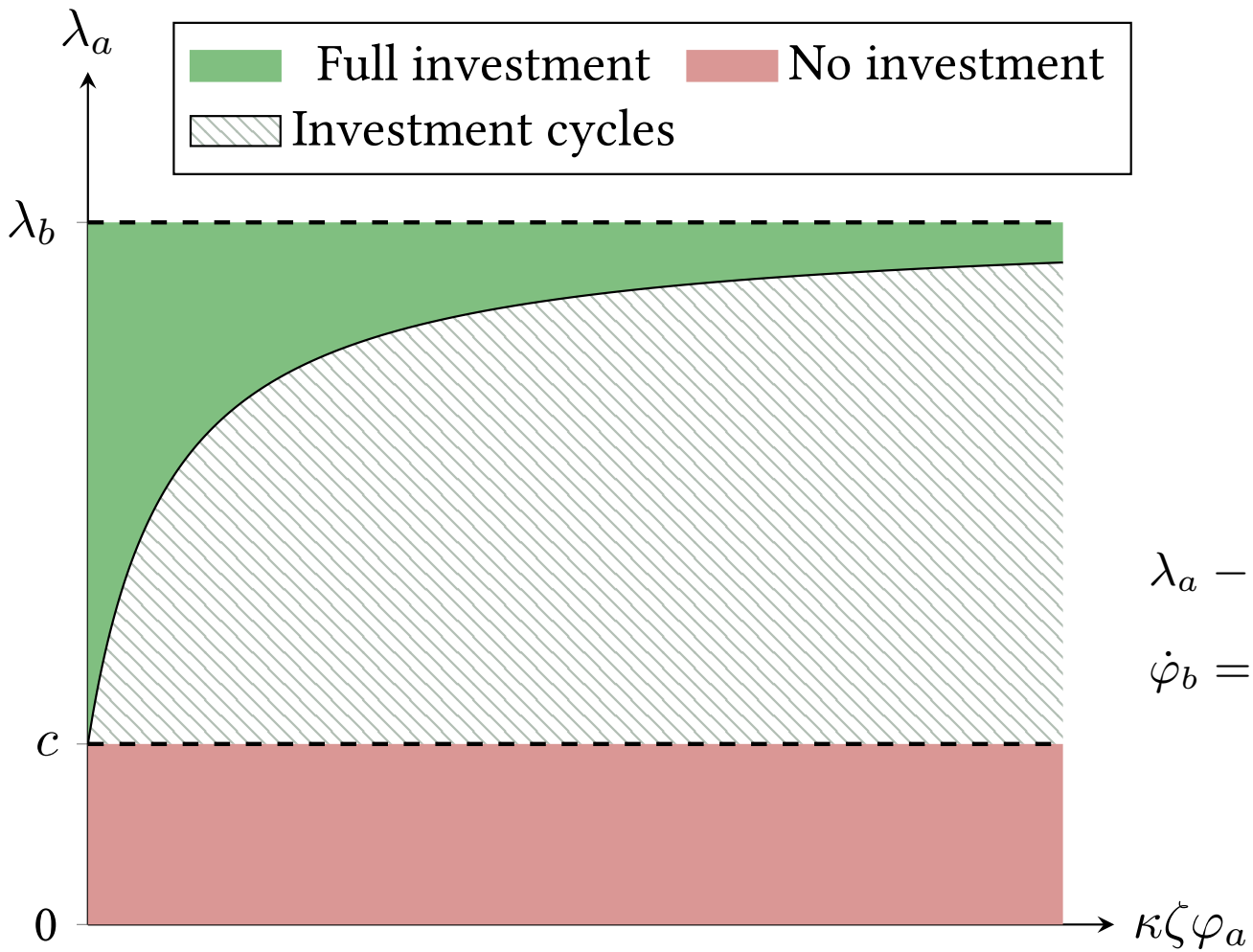
Figure 4: Equilibrium feedbacks with endogenous mass of projects

- Recall that investment in early stage project takes place iff

$$\lambda_a - c > \frac{\kappa\varphi_b(\lambda_b - c)}{\kappa\varphi_b + r + \lambda_b}$$

$$= \text{Opportunity cost of forgoing the option to wait for } b\text{-projects}$$

Spillover externality



$$\lambda_a - c > \overbrace{\frac{\kappa\varphi_b(\lambda_b - c)}{\lambda_b + r + \kappa\varphi_b}}^{\text{opportunity cost of forgoing the wait}}$$

$$\dot{\varphi}_b = \underbrace{\zeta\lambda_a m_a(1)}_{\text{inflow}} - \underbrace{\kappa\varphi_b n(1)}_{\text{outflow}}$$

Figure 5: Investment regimes under endogenous (φ_a, φ_b)

Decentralizing the social optimum I

- What would a *hypothetical benevolent planner* do? $\rightsquigarrow$ benchmark
- What are the planner's instruments? At best, the choice of matching sets $\rightarrow \{(\chi_a(\pi), \chi_b(\pi)) : \pi \in [0,1]\}$.
- Social optimum:

$$\boxed{\max_{\chi} \int_0^{\infty} e^{-rt} \sum_{q \in \{a,b\}} (\lambda_q \pi - c) M_q(\mathrm{d}\pi) \mathrm{d}t}$$

subject to law of motions for $\{m_q(\pi), n(\pi), \varphi_b\}$ dist. law of motion

Decentralizing the social optimum II

- Social marginal value functions (to be compared with private marginal value functions):

$$rv_*(\pi, q) = -c + \lambda_q \pi \left[1 + w_*(1) - v_*(\pi, q)\right] - \lambda_q \pi (1-\pi) \partial_\pi v_*(\pi, q) + \boxed{\rho \zeta \lambda_a \pi 1_{\{q=a\}}}$$

$$rw_*(\pi) = \sum_q \kappa \varphi_q \left[v_*(\pi, q) - w_*(\pi)\right] \chi_q^*(\pi) - \boxed{\rho \kappa \varphi_b \chi_b^*(\pi)}$$

- Optimal transfers (conceptual, not necessarily implementable from the public finance viewpoint)
  - A tax on unmatched investors that choose to invest in late stage projects = $\rho \kappa \varphi_b$
  - A subsidy to investors matched with early stage projects = $\rho \zeta \lambda_a \pi$

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Reputation and deal flow

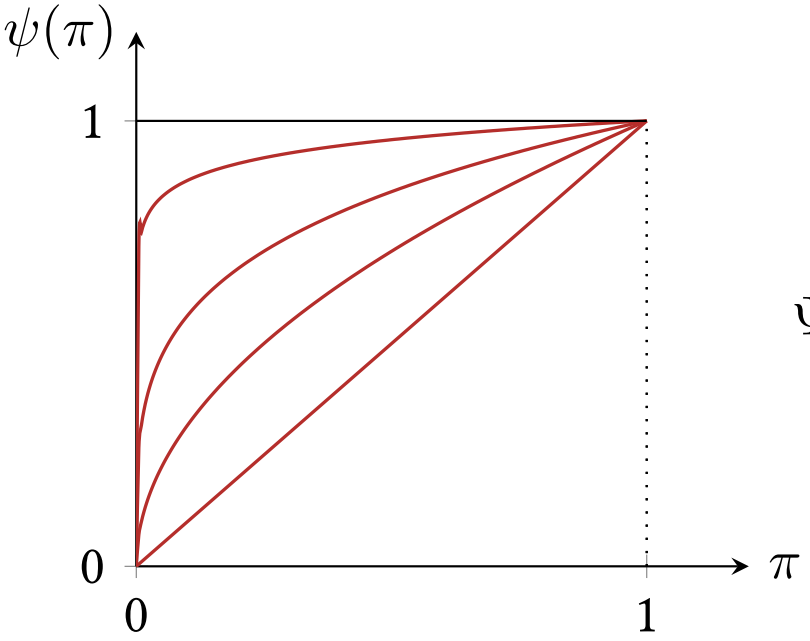
- Ample evidence that VCs reputation impact their deal flow e.g (Gompers 1996), (Sorenson and Stuart 2001), (Hsu 2004), (Nanda, Samila and Sorenson 2020).
- In the survey paper (Gompers, Gornall, Kaplan and Strebulaev 2020): 23% of VC responded the deal flow is the *most important* factor behind the success.
  - sub-sample of low IPO rate investors $\rightarrow$ 31%
  - sub-sample of high IPO rate investors $\rightarrow$ 19%
- Aggregate view: higher deal flow for more reputable ones $\Rightarrow$ lower deal flow for less reputable ones.

Reputational externality in the meeting rate function

- Suppose there is only one group of projects, i.e $\cancel{\lambda_a}$ and $\lambda = \lambda_b$.
- So far, uniform meeting rate $\Rightarrow \kappa\varphi \mathrm{d}t$. Now, reputational weighting, i.e:

$$\kappa\varphi \frac{\psi(\pi)}{\int \psi(\pi)\mathrm{d}F_\infty(\pi)}\mathrm{d}t = \kappa\varphi \frac{\psi(\pi)}{\underbrace{\mathsf{E}[\psi(\boldsymbol{\pi}_\infty)]}_{\mu:=}}\mathrm{d}t$$

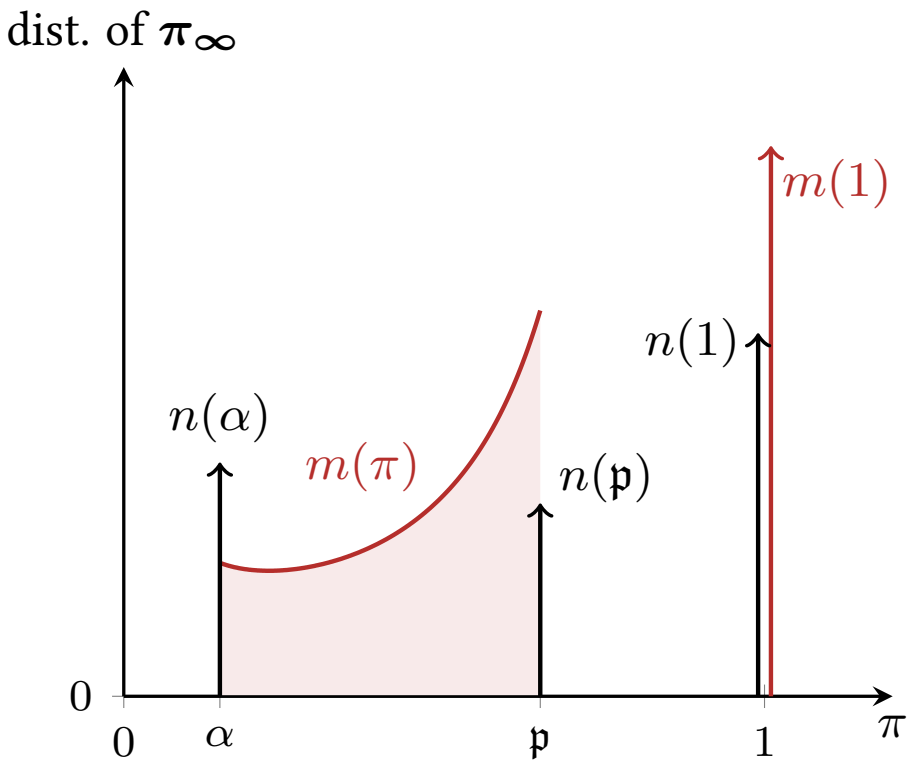
- Stationary economy $\rightarrow$ let investors exogenously enter and leave at the rate δ.



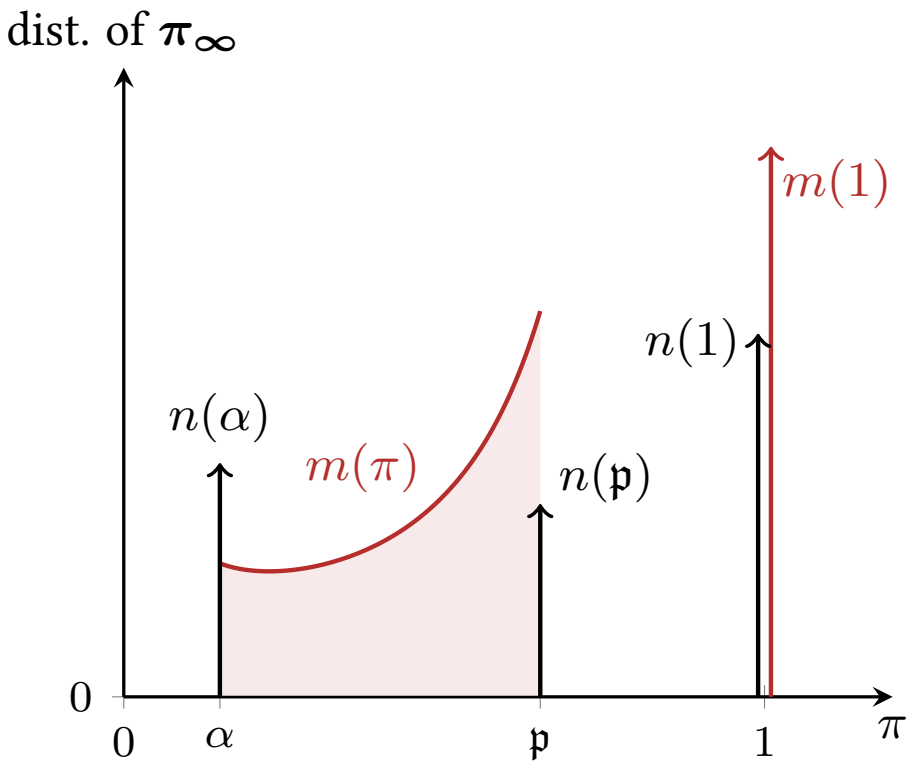
admissible space of weight functions

$$\Psi := \{\psi : [0,1] \rightarrow [0,1] | \\ \psi(0) = 0, \psi(1) = 1, \psi' \geq 0, \psi'' \leq 0\}$$

(Symmetric) equilibrium vs. the optimum outcome



(Symmetric) equilibrium vs. the optimum outcome

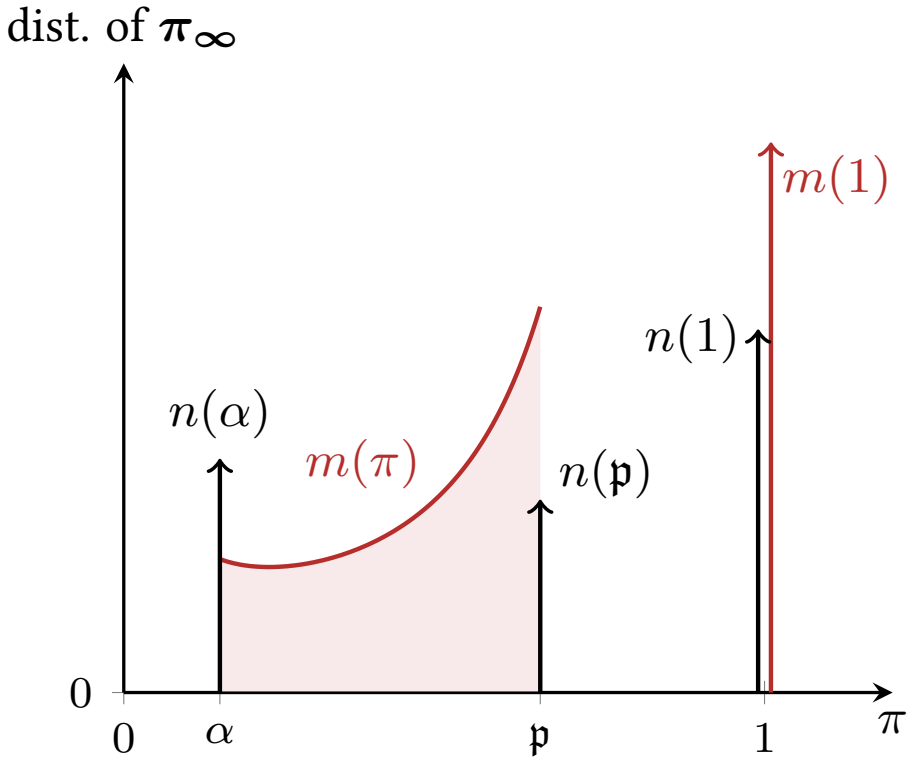


Properties of *symmetric* equilibrium

- cond (1) steady state μ:
  $\mu = \mathsf{E}\left[\psi(\boldsymbol{\pi}_\infty)\right] = \mathsf{M}(\mu, \alpha)$.
- cond (2) endogenous separation:
  $\alpha = \frac{c}{\lambda(1+w(1))} = \mathsf{A}(w(1))$
- cond (3) endogenous rep. value:
  $w(1) = \frac{(r+\delta)^{-1}\kappa\varphi/\mu}{r+\delta+\lambda+\kappa\varphi/\mu}(\lambda - c) = \mathsf{W}(\mu)$.

Fixed-point $\boxed{\mu_e = \mathsf{M}\left(\mu_e, \overbrace{\mathsf{A} \circ \mathsf{W}(\mu_e)}^{\alpha_e}\right)}$

(Symmetric) equilibrium vs. the optimum outcome



Properties of *symmetric* equilibrium

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- cond (3) endogenous rep. value:
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Fixed-point $\boxed{\mu_e = \mathsf{M}\left(\mu_e, \overbrace{\mathsf{A} \circ \mathsf{W}(\mu_e)}^{\alpha_e}\right)}$

Planner's optimum:

$$\max_{\alpha} rS(\mu, \alpha) = (\lambda - c)m(1) + \int_{\alpha}^{\mathfrak{p}} (\lambda\pi - c)m(\pi)\mathrm{d}\pi$$

$$\text{subject to} \quad \mu = \mathsf{M}(\mu, \alpha)$$

Decentralized vs. centralized outcomes

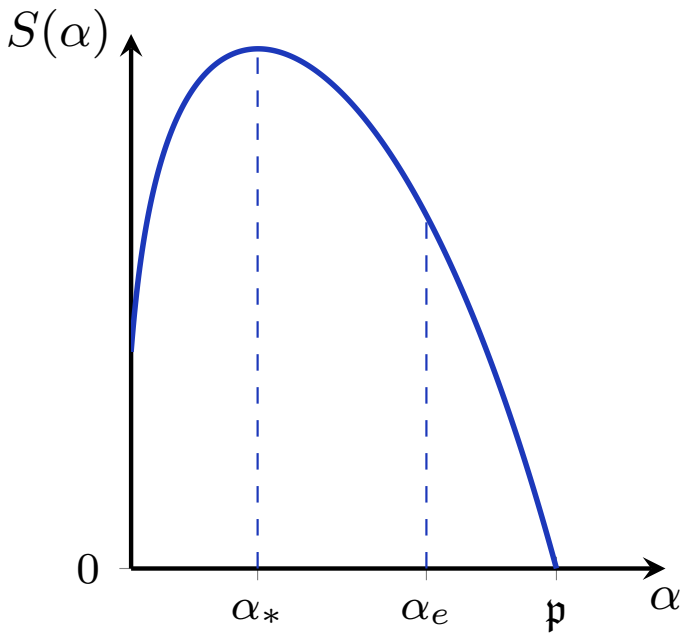
Proposition 2: Equilibrium vs planner outcomes

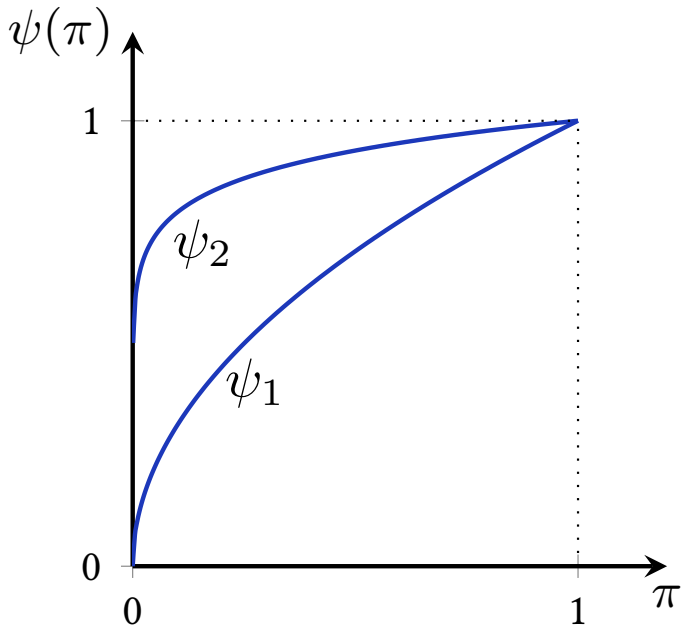
In the described economy with reputational externality,

(i) there always exists a symmetric equilibrium with $0 < \alpha_e < \mathfrak{p}$.

(ii) In particular, a local *reduction* in the termination point α_e increases the social surplus.

(iii) Comparative statics: $\psi \uparrow \Rightarrow w(1) \downarrow$, $\mu_e \uparrow$ and $\alpha_e \uparrow$





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Robust extension

- Economy with short-lived investors, where individuals are subject to exogenous exits and exogenous birth.
- General type space for project qualities, $q \sim \phi(\mathrm{d}q)$, and *inconclusive* breakthroughs, i.e $\lambda_q(\theta)$ not necessarily equals $\theta\lambda_q$:
  - $\lambda_q(\theta)$ is increasing in q.
  - Increasing differences, i.e for every $q'' > q'$:

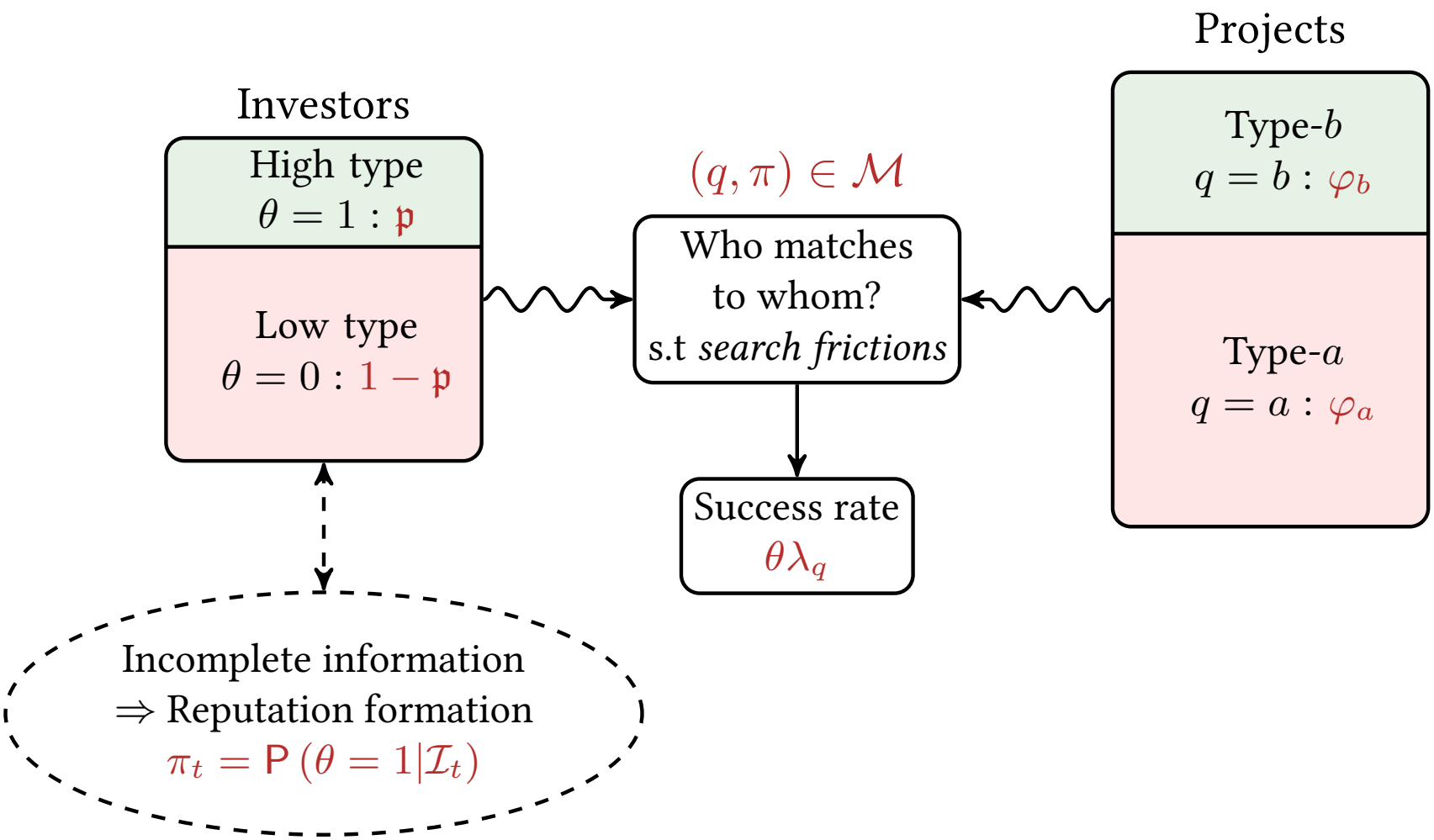
$$\lambda_{q''}(H) - \lambda_{q''}(L) \geq \lambda_{q'}(H) - \lambda_{q'}(L)$$

proof sketch

Conclusion

- Characterization of matching sets in the unique stationary equilibrium with *increasing* value functions in reputation:
  - Higher tolerance for failure in more reputable VCs.
  - Higher search frictions could save the market from breakdown when there is spillover from early- to late-stage projects.
- In presence of reputational externality:
  - Early termination of startups.
  - Size of active high-ability investors becomes inefficiently small.
  - Strengthening the reputational externality ($\psi \uparrow$) leads to lower tolerance for failure ($\alpha_e \uparrow$).

Model diagram



Back to the model

Non-recursive form of the value functions

- Matching value function:

$$v(\pi, q) = \sup_{\tau} \left\{ \mathsf{E}\left[e^{-r\sigma} - c\int_0^{\sigma} e^{-rt}\mathrm{d}t + e^{-r\sigma} w(\pi_\sigma); \sigma \leq \tau \right] \right.$$
$$\left. + \mathsf{E}\left[-c\int_0^{\tau} e^{-rt}\mathrm{d}t + e^{-r\tau} w(\pi_\tau); \sigma > \tau \right] \right\}$$

- Reputation value function:

$$w(\pi) = \sup_{\mathcal{M}} \left\{ \mathsf{E}\left[e^{-r \min \tau_q} v(\pi, \arg\min \tau_q) \right] : (q, \pi) \in \mathcal{M} \right\}$$

Back to the recursive forms

Distributional law of motions

$$\dot{m}_q(\pi) = -\underbrace{\lambda_q \pi m_q(\pi)}_{\substack{\text{successful projects}\\ \text{outflow}}} + \underbrace{\kappa \varphi_q n(\pi) \chi_q(\pi)}_{\substack{\text{inflow of}\\ \text{recent matches}}} + \underbrace{\lambda_q \partial_\pi \left(\pi(1-\pi) m_q(\pi)\right)}_{\text{net learning inflow}}$$

$$\dot{n}(\pi) = -\underbrace{\sum_q \kappa \varphi_q n(\pi) \chi_q(\pi)}_{\substack{\text{outflow of}\\ \text{recent matches}}}$$

$$\dot{\varphi}_b = \underbrace{\zeta \lambda_a \left(m_a(1) + \int \pi m_a(\pi) \mathrm{d}\pi \right)}_{\substack{\text{spillover from successful early}\\ \text{to late stage projects}}}$$

$$- \underbrace{\kappa \varphi_b \left(n(1) \chi_b(1) + \int n(\pi) \chi_b(\pi) \mathrm{d}\pi \right)}_{\substack{\text{outflow due to}\\ \text{recent partnerships}}}.$$

Back to planner's problem

Proof sketch for the *general* type space I

- Projects' type space $q \sim \phi(\mathrm{d}q)$. Assume $\lambda_L(q) \leq \lambda_H(q) \leq \boldsymbol{\lambda}$.
- Stopping time problem:

- Fixed-point mapping $w = \mathcal{A}w$:

$$[\mathcal{A}w](\pi) = \sup\left\{\frac{\int_B [\mathbb{T}_q w](\pi)\ \phi(\mathrm{d}q)}{1 + \frac{\kappa}{r}\phi(B)} : B \subset \mathsf{Supp}(\phi)\right\}$$

- Let $L^1[0,1]$ be the underlying space for w, endowed by the partial order $\succsim$ induced by $L^1_+[0,1]$.
- Properties of $\mathcal{A}$:
  - Positivity: $\mathcal{A}w \succsim \mathbf{0}$, for every $w \succsim \mathbf{0}$.
  - Monotonicity: if $w_2 \succsim w_1 \Rightarrow \mathcal{A}w_2 \succsim \mathcal{A}w_1$.
  - Every fixed-point of $\mathcal{A}$ is *order-bounded* by $\boldsymbol{\lambda}/r$.

Proof sketch for the *general* type space II

- Let $\langle \mathbf{0}, \boldsymbol{\lambda}/r \rangle := \left\{ f \in L^1_+[0,1] : \mathbf{0} \precsim f \precsim \boldsymbol{\lambda}/r \right\}$ be the order interval, then $\mathcal{A} : \langle \mathbf{0}, \boldsymbol{\lambda}/r \rangle \to \langle \mathbf{0}, \boldsymbol{\lambda}/r \rangle$ 🔔 Knaster–Tarski theorem 🔔
- Use a coupling argument along with the Strassen's theorem to show $\mathcal{A}$ maps increasing functions to increasing functions.
- Construct the sequence $w_0 = \mathbf{0}$ and $w_n = \mathcal{A} w_{n-1} \Rightarrow w_n \uparrow w_\infty$ pointwise, where w_∞ is increasing.
- Showing the L^1 continuity: $\|\mathcal{A} w_n - \mathcal{A} w_\infty\|_1 \to 0$, therefore

$$w_\infty = \mathcal{A} w_\infty.$$

- Using the complementarity in $\lambda_q(\theta)$ to show $\exists\, \alpha$, the lowest boundary point of $\mathcal{M}_b$, where $b = \sup\{\mathrm{Supp}(\phi)\}$.

back